Diagnosing Why a YouTube Video Underperforms

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1. Problem Statement

Content creators have no reliable way to know, before or after publishing, why a specific video underperformed relative to what succeeds in its own category. Existing engagement research Wu et al. (2018) shows that a video's eventual performance can be predicted from its content and channel properties alone, even before it accumulates any view history, but stops short of explaining *why* a specific video is likely to underperform. We address this gap with YT-Diag: given any public YouTube video, the system predicts its likely performance within its own content niche and produces a short, ranked, human-readable explanation of the content features most likely responsible. Our approach combines three deep learning modalities: a pretrained text transformer over the title, description, and transcript; a pretrained vision backbone over the thumbnail; and a lightweight temporal transformer over frames sampled across the full video Abu-El-Haija et al. (2016), fused into a single model trained to predict a within-category viral/non-viral label, with an attention-based attribution layer surfacing the specific features driving each prediction.

2. Existing Methods

Consumer tools such as VidIQ, TubeBuddy, and newer entrants like OutlierKit and Virlo already offer creators feedback on why their content might not be performing, but rely on metadata and SEO-style heuristics (keyword scoring, upload timing, tag suggestions) rather than analyzing the video's actual visual and spoken content. Academically, Wu et al. (2018) established that engagement can be measured and predicted from content and channel metadata alone in a cold-start setting, at a scale of 5.3 million videos, and introduced "relative engagement" as a stable, comparable performance metric. More recent work Rajaram and Manchanda (2020) has shown that fusing text, audio, and video-image signals through an interpretable attention-based deep learning framework improves both predictive accuracy and the ability to explain which features drive engagement, though this work is applied to influencer marketing videos broadly rather than packaged as a per-video diagnostic tool. Our contribution is the combination of genuine content-level multimodal analysis, a validated attribution layer, and a category-conditioned comparison, so that a small channel's video is judged against realistic peers rather than global outliers, applied specifically to the practical question of why a given video underperformed.

3. Data Description

We build our own labeled corpus using the official YouTube Data API v3. Scope: four content categories (comedy/skits, tutorials & how-to, vlogs, and product reviews), chosen because virality drivers differ meaningfully by category. Target size: approximately 500 videos per category (2,000 videos total), split between videos flagged as "viral" (top quartile of views-per-day-since-upload, normalized by the channel's subscriber count, within category) and "typical" videos from similarly sized channels in the same category. Per video we collect: 12 structured metadata features (duration, publish hour and day-of-week, tag count, title and description length, category ID, channel subscriber count and age, plus view/like/-comment counts used only to construct the label, never as model input); the thumbnail image (1280x720 JPG, encoded via a pretrained vision backbone); 16–24 frames sampled uniformly across the full video (encoded with the same vision backbone, aggregated with a lightweight temporal transformer); the video transcript (YouTube auto-captions where available, on an estimated 70–80% of videos, with Whisper as fallback transcription); and the title and description text (encoded via a pretrained text transformer).

4. Methodology

We adapt, rather than adopt wholesale, the open-source codebase accompanying Wu et al. (2018) (`https://github.com/avalanchesiqi/youtube-engagement`), which defines the relative engagement metric and provides content/channel-feature baselines we reuse as our non-deep-learning comparison point. We extend this with a multimodal deep learning layer: a pretrained vision backbone over the thumbnail, a pretrained text transformer over the title and transcript, and a temporal transformer over sampled video frames following the frame-sampling-plus-pretrained-CNN methodology used at scale in Abu-El-Haija et al. (2016), fused via a small cross-attention or concatenation layer and trained end-to-end to predict the within-category viral/non-viral label. An attention- or ablation-based attribution layer, inspired by Rajaram and Manchanda (2020), surfaces which specific elements likely depressed a given video's predicted performance. Baselines: logistic regression and XGBoost on structured metadata alone, to test whether the deep multimodal signal adds real predictive value beyond scheduling and format. Key challenges: obtaining full video files for frame sampling sits outside YouTube's terms of service unless a video is Creative Commons-licensed, so this component will either be scoped to CC-licensed videos or proceed with the limitation disclosed; "virality" requires a defensible, stated threshold, since no universal definition exists; a large share of real-world virality is driven by factors outside the content itself (platform promotion, timing, luck), capping achievable accuracy regardless of model quality, which we report honestly rather than overclaim; and fusing four heterogeneous signal types into one coherently trained model, validated against a non-deep-learning baseline, within a one-month timeline.

References

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